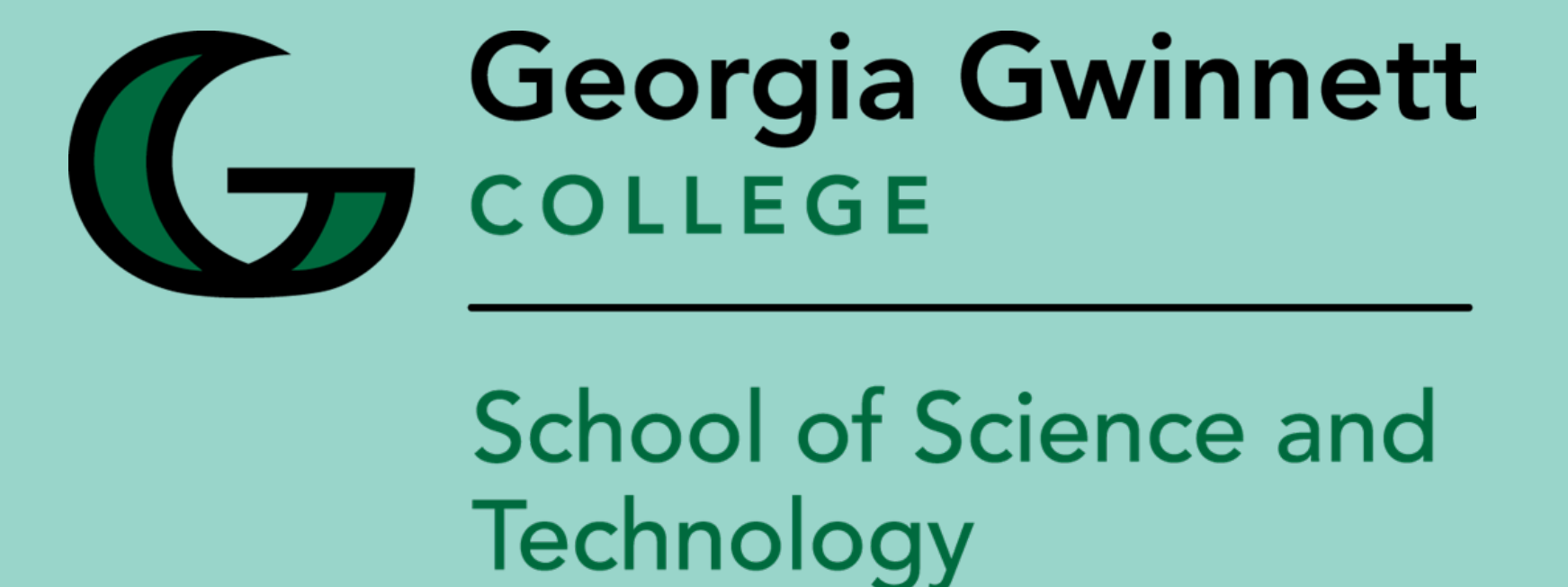


TEACHING SIGN LANGUAGE WITH AI

Advisors: Dr. Wei Jin, Dr. Xu Xin By: Joey Chen, Keren Zhang, Ivana Hernandez Fall 2025

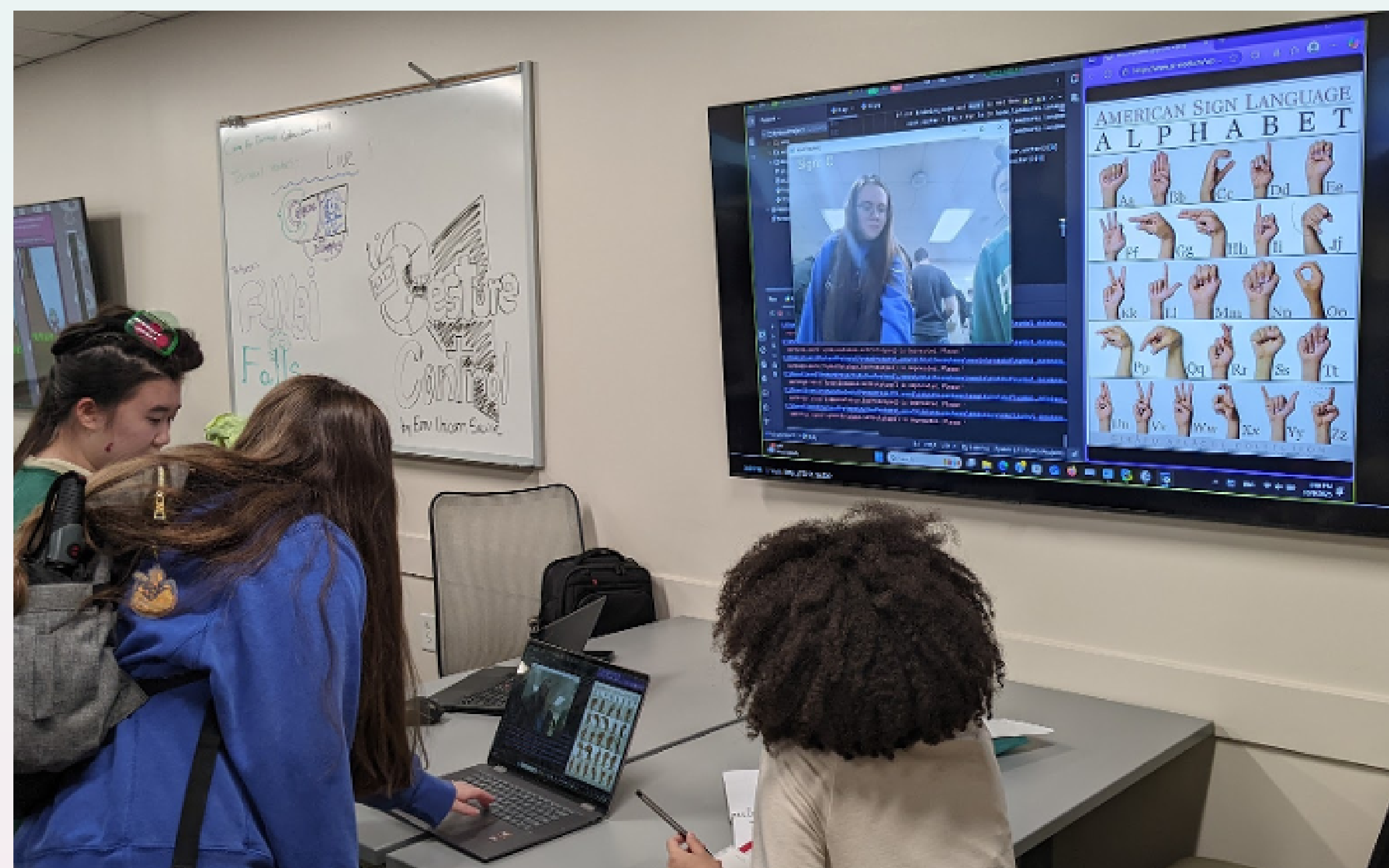


ABOUT TAP

The **Technology Ambassador Program (TAP)** at GGC prioritizes simplifying technology to the common eye. Students choose and work on a project, developing both technical skills and soft skills. Anybody can, and is encouraged to, enroll! ☺

GOALS

- To showcase a tangible, real-world use of AI, with learning sign language as the bridge
- To demystify AI and machine learning



Team Emu Unicorn Sauce

GESTURE CONTROL

Gesture Control is a program based off *Mediapipe's* hand landmark code. The user can select a letter to sign, and it grades them on their accuracy as the hand comes into frame.

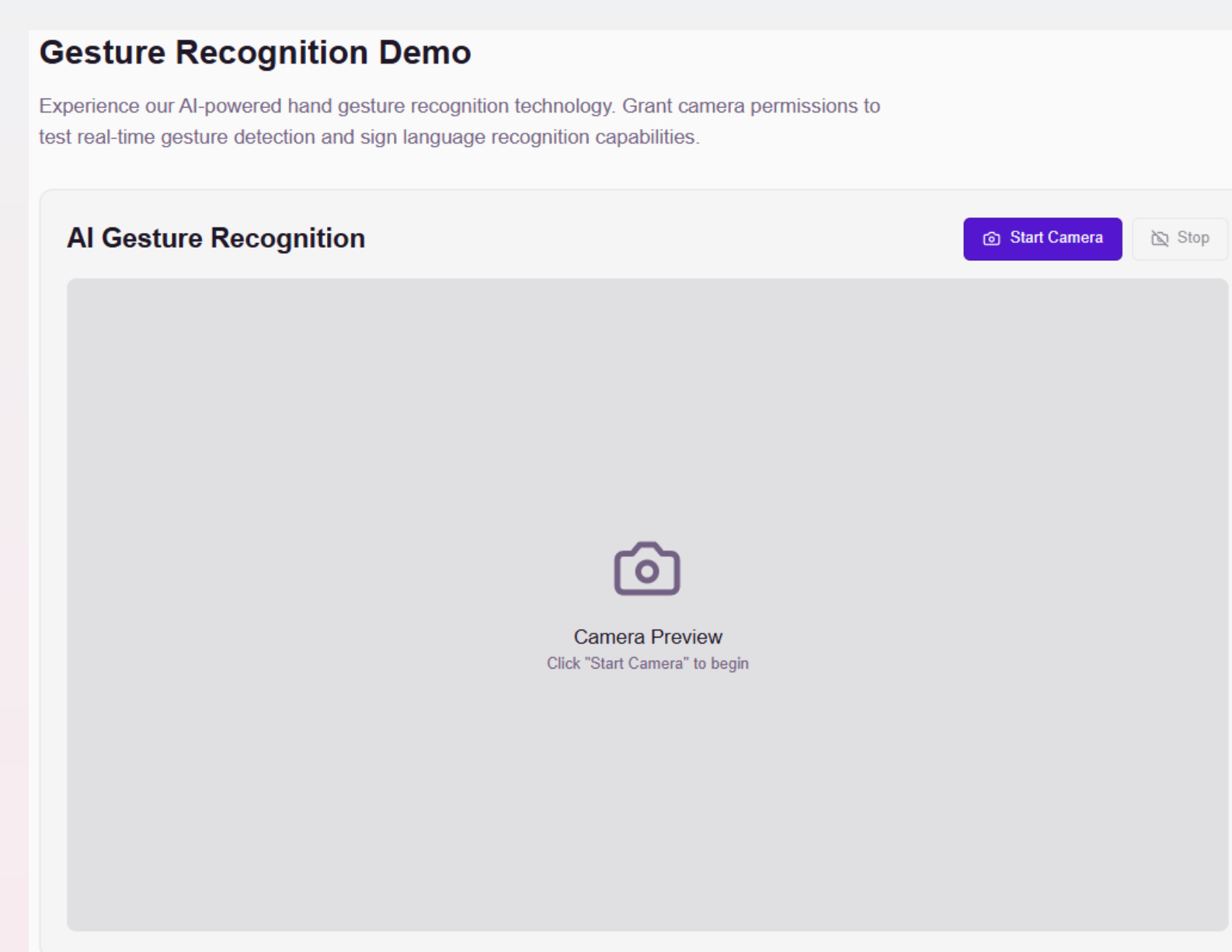
- The data the AI was trained on is stored in a CSV file



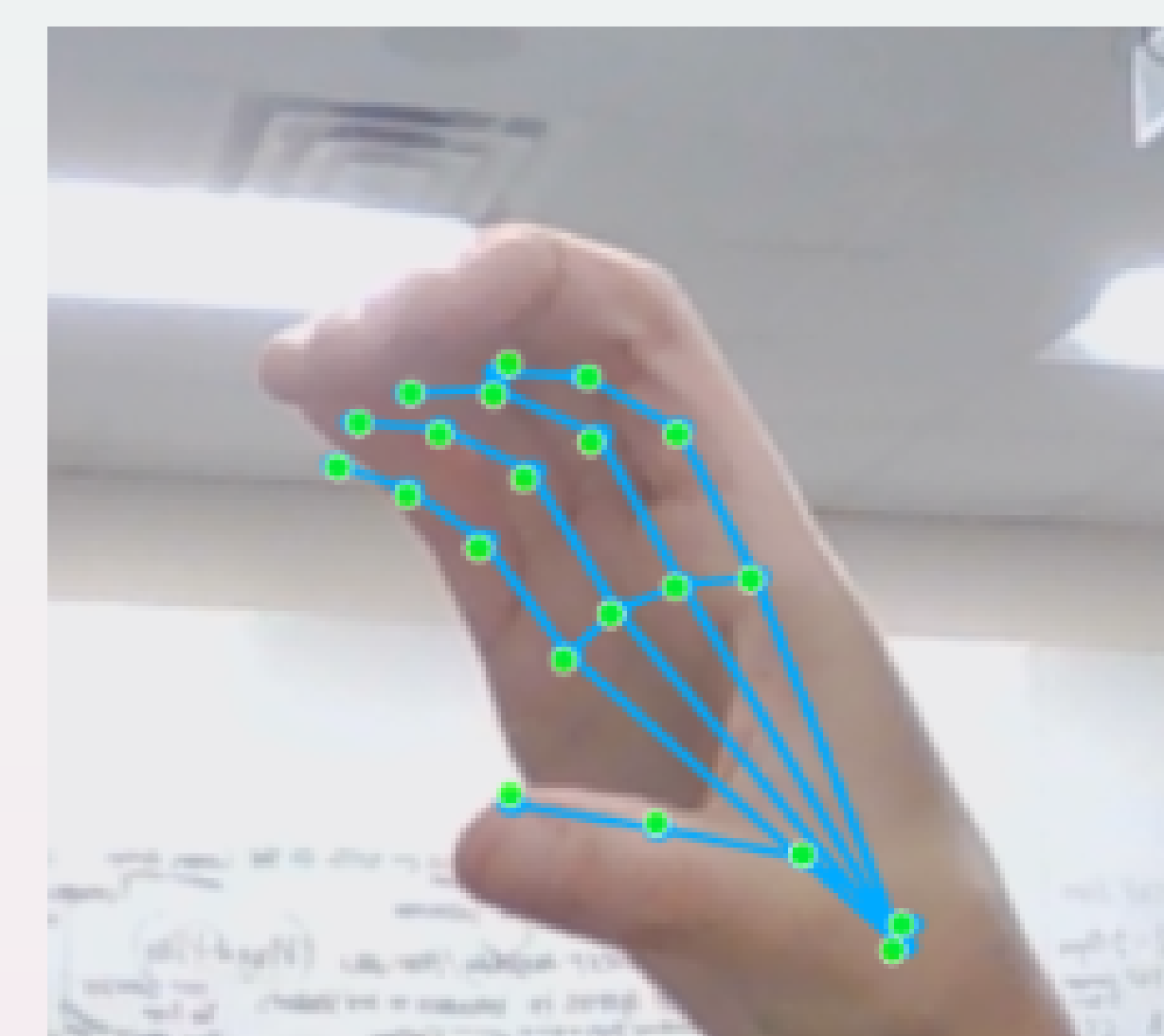
WORKSHOPS

Students will access the webpage by link on a device, which will then prompt for the user's camera. The user can choose which letter to sign, with tips at the bottom of the page unique to each letter.

- Participants will take a pre and post survey to quantify the effectiveness of the workshops



(Image of the website before webcam access is granted)



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RESULTS

Students were quizzed on their previous on their confidence and experience in the areas of programming and IT in general. The Pre/Post surveys show improvement in attitude in these fields after the workshop. Students showed mixed results when it came to answering content-specific questions, conflating the AI used for *Gesture Control* and AI in general.

